

Designed Interaction for World-Model Learning

An exploration policy that collects what the model does not yet know

Research Proposal — v1.0

July 2026

Abstract

Action-conditioned world models are now good enough to train and evaluate robot policies, but the data used to train *them* is collected by hand-tuned heuristics: scripted random motion, human play data, and manually constrained action spaces. We formulate world-model data collection as an optimization over the data-generating distribution and give an algorithm for it: a residual policy on top of a frozen task policy, trained entirely in the world model’s imagination against an information-gain reward derived from a score-function disagreement between ensemble members. The reward is destroyed by its own success, which forces iteration and yields a free saturation diagnostic: when the residual can no longer find informative interactions, more data of that kind will not help. We argue the central claim conditionally — design helps exactly when the naive data distribution is rank-deficient in the directions that matter — and we treat our own prior negative result as the motivating measurement rather than a failure.

1 Problem Statement

Let f_ϕ be an action-conditioned world model trained by minimizing a per-transition loss $\ell(\phi; z)$ on transitions $z = (x, u, x')$. A data-generating policy π induces an occupancy measure μ_π over transitions, constrained to the reachable set

$$\mathcal{D} = \{\mu \geq 0 : \mathcal{L}^* \mu = \nu_0\}, \quad (\mathcal{L}^* \mu)(B) = \mu(B \times U) - \gamma \int P(B \mid x, u) \mu(dx du), \quad (1)$$

the Bellman flow (discrete Liouville) constraint. Training yields $\phi^*(\mu) = \arg \min_\phi \langle \ell(\phi; \cdot), \mu \rangle$. Let $U(\phi)$ be a downstream utility — the task score of a control policy trained inside f_ϕ . Given a budget of B episodes:

$$\boxed{\mu^* = \arg \max_{\mu \in \mathcal{D}, \|\mu\| \leq B} U(\phi^*(\mu))}. \quad (2)$$

The question is not how much data to collect but *which*. We ask when that question matters, how to answer it algorithmically, and how to certify when it does not matter.

Scope. We assume no physics simulator and no parametric physics vector: ϕ is the world model’s own parameters. We do not assume a learned exploration policy is necessary — whether it is, is one of the questions (§4.8). We target manipulation with contact, where object dynamics are not determined by the robot’s commanded trajectory.

2 Motivation

2.1 Data collection for world models is currently hand-tuned

Recent work establishes that action-conditioned video world models can serve as training and evaluation environments: policies trained purely on world-model rollouts match policies trained on equal amounts of real data, with simulated-to-real task-score correlations of 0.85–0.99. The data used to train those world models, however, is specified procedurally — a scripted random policy “to ensure sufficient data coverage,” roughly 600 human play episodes per task (about six person-hours each), and, reported as an implementation detail, a hand-constrained action space (e.g. restricting the end-effector to planar motion) to make the task learnable at all.

That last item is manual dimensionality reduction of what the world model must identify: an informal instance of the design problem we formalize. The stated open question in that line of work is how world-model performance *scales* with data quantity. We address the complementary question of *composition*.

2.2 What “understands physics” means

Three candidate definitions are not equivalent:

Definition	Formal	Problem
fits the data	$\mathbb{E}_{\mu_{\text{data}}}[\ell]$ small	memorizes appearance (this is PSNR)
predicts held-out frames	same, held-out split	averages over <i>behavior</i> actions
responds to interventions	$\sup_u d(f_\phi, P)$ small	—

Physics understanding is a claim about $\text{do}(u)$, not $p(u)$: the model understands the physics at x if it predicts correctly for actions never taken at x . Hence the target

$$\mathcal{J}(\phi) = \mathbb{E}_{x \sim \rho} \left[\sup_{u \in \mathcal{U}_\epsilon} d(f_\phi(\cdot | x, u), P(\cdot | x, u)) \right]. \quad (3)$$

The sup is the entire content. Standard training minimizes an *average* over $u \sim \pi_{\text{behavior}}$; nothing in that objective controls the worst case, and physics is exactly the worst case — the response to the action you did not take. Parameterizing the inner sup with a policy gives

$$\min_{\phi} \max_{\rho} \mathbb{E}_{\mu_{\pi_b + \Delta \rho}} [\ell(\phi; z)] \quad \text{s.t.} \quad \text{KL}(\pi_\rho \| \pi_b) \leq \epsilon. \quad (4)$$

The residual policy is the inner maximization. This is not an exploration heuristic; it is a sample-based inner maximization for a worst-case objective. Passive and adversarial training optimize *different functionals*, not the same functional at different efficiencies.

The KL ball makes the sup well-posed. Unconstrained, it is dominated by degenerate actions — observed empirically as “camera-thrashing” reward hacking at a weak anchor in prior work.

2.3 Two structural facts from the occupancy view

(a) Prediction-error acquisition has a degenerate optimum. Maximizing prediction error is $\max_{\mu} \langle \ell, \mu \rangle$: *linear* over a polytope. LP optima are vertices, and vertices of $\mathcal{D}$ are deterministic policies. The exact optimum of curiosity is therefore “repeat the single highest-loss interaction until the budget is spent” — redundant data, rank-one information. This is not a tuning failure. It

is the continuous form of the top- K redundancy pathology known from BatchBALD and core-set selection.

Design objectives are concave composed with linear, $\Phi(M(\mu))$ with $M(\mu) = \langle gg^\top, \mu \rangle$, and have interior optima with spread support. The curvature is what prevents redundancy.

(b) Prediction error is the wrong quantity. Decomposing expected loss,

$$\mathbb{E}_\mu[\ell] = \underbrace{\text{tr}(\Sigma_\phi M(\mu))}_{\text{epistemic}} + \underbrace{\sigma_{\text{alea}}^2}_{\text{observation noise}} + \underbrace{b^2}_{\text{misspecification}}, \quad (5)$$

whereas expected information gain, under $p(x' | x, u, \phi) = \mathcal{N}(f_\phi, \sigma^2)$,

$$I(x'; \phi | x, u) \approx \frac{1}{2} \log \left(1 + \frac{\text{Var}_\phi[f_\phi]}{\sigma^2} \right), \quad (6)$$

depends only on the first term. Loss ranks on all three; information gain on one. This predicts the reward hacking observed in prior work and is our principal algorithmic difference from it.

2.4 Some quantities are unidentifiable at any sample size

An object sliding freely to rest decelerates at $a = \mu_{\text{fric}} g$ — independent of mass. So $\nabla_{(m, \mu_{\text{fric}})} a = (0, g)$, every sliding observation contributes a gradient along $(0, 1)$, and

$$M(\bar{\mu}_{\text{slide}}) \propto \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{mass} = C = (1, 0) \notin \text{range } M. \quad (7)$$

A thousand hours of sliding determines nothing about mass. Breaking the degeneracy requires a known impulse or a collision with a known-mass body. This is a statement about invertibility, not efficiency, and it makes the necessity claim conditional:

$$\text{design necessary} \iff C \notin \text{range } M(\bar{\mu}_0) \text{ but } C \in \text{range } M(\bar{\mu}^*) \text{ for some } \bar{\mu}^* \in \mathcal{D}. \quad (8)$$

2.5 Preliminary negative result, and why it was predictable

We ran a two-arm pool-based active-learning experiment on a single LIBERO scene: strategic (top- K highest teacher-forced denoising loss) versus random, matched at 50k training steps. Strategic was *worse* (PSNR 24.23 vs 24.60, $t = -2.46$, $p \approx 0.017$; SSIM tied), with superimposable training curves and an identical loss floor. Three confounds, all predicted by §2.3:

1. The candidate pool was **reach** (grasp — the object then moves rigidly with the gripper, hence is a deterministic function of the action the model already receives), **perturb** (demo trajectory plus noise), and **random** (mostly free space). **No interaction in the pool excited object dynamics independent of the commanded trajectory.** The pool was rank-deficient in the parameter directions of interest.
2. Top- K on raw loss is exactly failure mode (a).
3. The loss floor was misspecification plus reconstruction error — terms invisible to any information objective and unfixable by data selection.

We treat this as the motivating measurement: it is what the framework predicts in the regime where design cannot help. Reproducing the tie *as a prediction* is part of the contribution.

3 Related Work

Design over occupancy measures. The LP formulation of MDPs originates with d’Epenoux, De Ghellinck and Manne (1960) and is developed functional-analytically by Hernández-Lerma & Lasserre (1996, 1999) and via Borkar’s convex-analytic approach (1988). Optimizing a *concave* function of the occupancy measure is the convex-MDP setting of Zahavy et al. (NeurIPS 2021), who show such goals are not expressible as stationary rewards and give a Fenchel min–max reformulation. For model estimation specifically: FW-Modest (Tarbouriech et al., 2020) and κ -Explorer (2026), both Frank–Wolfe over state-action occupancy measures. *We use this machinery; we do not claim it.*

Task-directed design. Wagenmaker, Simchowicz & Jamieson (ICML 2021) show task completion reduces to maximizing a task-specific linear functional of the Fisher information matrix, recovering classical A -/ c -optimal design; DynamicOED (2023) extends this to nonlinear systems. Independently, decision-aware and value-equivalent model learning (Farahmand et al. 2017; Farahmand 2018; Grimm et al. 2020, 2021; Abachi et al. 2020) argues model error should be weighted by its effect on downstream value.

Fisher-information exploration in robotics. ASID (Mommel et al., ICLR 2024) trains an exploration policy in simulation maximizing Fisher information about simulator parameters, then runs system identification and trains a task policy. Sathyanarayan & Abraham (RSS 2025) derive a contact-aware Fisher information measure with emergent tactile behaviors for mass, friction, stiffness and shape; Vantilborgh et al. (2025) rank contact data by informativeness and find fewer informative samples beat more data. QOED (2026) isolates identifiable eigendirections of the FIM. **All presume a parametric simulator with an explicit physics vector and a likelihood over it.**

Exploration for world-model learning. Plan2Explore (Sekar et al., ICML 2020) is the direct ancestor of our method: latent ensemble disagreement as reward, actor trained in imagination inside a learned world model. Pathak et al. (2019) use the same signal without a world model. SENSEI (2025) guides exploration with foundation-model semantics and reports that task-free Plan2Explore does not consistently beat learning from scratch. These are *task-agnostic*; we anchor to a task policy.

Adversarial curricula for world models. PROWL (arXiv:2605.18803, May 2026) is the closest prior work and largely coincides with our formulation: a KL-anchored adversarial policy trained with PPO against a diffusion video world model, with the world model fine-tuned on discovered failures, expressed as an asymmetric two-player game. Differences that remain, in order of importance:

1. **PROWL’s adversary requires environment rollouts for every policy update**, because its signal is realized regret against ground-truth rollouts. Ours is optimized *entirely in imagination*. In Minecraft that is cheap; on a physical robot it is the dominant cost.
2. Signal: realized regret (contaminated by the two irreducible terms in (5)) versus epistemic disagreement. Their observed camera-thrashing reward hacking is the predicted consequence; their fixes — a stronger KL anchor and swapping pixel differences for optical flow — are patches on the integrand.

3. Domain: game video versus robot manipulation with contact.
4. Metric: world-model fidelity (latent regret, action-follow score, LPIPS) versus downstream policy task score.
5. No saturation certificate.

We adopt their prioritized-replay learning-progress term and treat PROWL as the primary baseline.

Data selection. RHO-LOSS (Mindermann et al., 2022) shows greedy max-loss selects noisy or non-learnable points and proposes reducible loss. BatchBALD (Kirsch et al., 2019), core-set (Sener & Savarese, 2018) and BADGE (Ash et al., 2020) show top- K on any per-point score acquires redundant batches and can underperform random. Influence functions (Koh & Liang, 2017) are known to be fragile for deep non-convex models (Basu, Pope & Feizi, 2021).

Residual policies. Residual RL on a frozen base policy is mature: ResFiT, Policy Decorator, TRANSIC, ZPRL, LodeStar, including the stabilizing devices we adopt (bounded residual magnitude, progressive exploration schedules). **In all of this work the residual exists to improve task performance**, not as a data-collection mechanism for a separately trained model.

4 Method

4.1 Components

Symbol	Object	Trained on
f_ϕ	pixel world model (small, single forward pass per step)	$\mathcal{L}_{\text{WM}}$
$\{\phi_i\}_{i=1}^N$	ensemble members, frozen shared encoder	bootstrap, independent seeds
π_b	flow-matching BC policy	frozen forever
Δ_ρ	residual policy	r (never task reward)
V_ρ	chunk-level critic	advantage estimation
$\mathcal{B}$	replay buffer, weighted, never pruned	—

If Δ_ρ ever sees task reward, the method collapses back to Dreamer.

4.2 The score: information gain via score-function disagreement

The exact identity relating KL divergence to score differences across noise levels is

$$\text{KL}(p \parallel q) = \int_0^T \frac{1}{2} g(\sigma)^2 \mathbb{E}_{z_\sigma \sim p_\sigma} \left[\|\nabla \log p_\sigma - \nabla \log q_\sigma\|^2 \right] d\sigma, \quad (9)$$

the rigorous form of the gradient used by Distribution Matching Distillation, where the KL gradient is expressed as a difference of two score functions. Expected information gain (BALD) is the average KL from each posterior sample’s predictive to the mixture predictive,

$$I(z'; \phi \mid z, a, \mathcal{B}) = \mathbb{E}_{\phi \sim p(\phi \mid \mathcal{B})} \left[\text{KL}(p(z' \mid z, a, \phi) \parallel p(z' \mid z, a, \mathcal{B})) \right], \quad (10)$$

which combined with the identity above yields our reward:

$$r(z_k, a) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\sigma, \varepsilon} \left[w_\sigma \|v_{\phi_i}(z_\sigma, \sigma \mid z_k, a) - \bar{v}(z_\sigma, \sigma \mid z_k, a)\|_{\mathcal{O}}^2 \right] + \lambda_{\text{prog}} \Delta r_k^{\text{hist}} \quad (11)$$

with $z_\sigma = (1 - \sigma)z_k + \sigma\varepsilon$ and $\bar{v} = \frac{1}{N} \sum_i v_{\phi_i}$.

Five implementation requirements. Each is one line of code and the method fails without it.

1. **Velocity differences, not converted scores.** v and s are related by an affine, σ -dependent transform, so $\|v_i - \bar{v}\|^2 \propto \|s_i - \bar{s}\|^2$ up to a factor absorbed into w_σ . No conversion needed.
2. **Variance of predictions**, never of scalar losses. Variance of losses measures disagreement about *how wrong* members are, which is nearly uninformative.
3. **Object tokens only** ($\|\cdot\|_{\mathcal{O}}$). Project the object bounding-box corner stream to the latent token grid. Without this, disagreement is dominated by background, lighting and texture, and the residual optimizes wallpaper.
4. **Adaptive w_σ .** Scores diverge at low probability; DMD’s gradient-magnitude normalization outperformed the standard σ_t/α_t and σ_t^3/α_t schemes by 0.9 FID. Sweep σ_{max} : their own logs show this mattering more than most hyperparameters, since at high σ all members approach the same Gaussian prior and the difference vanishes.
5. **Learning-progress term.** Raw disagreement is high both where the model has not yet learned and where it *cannot* (capacity, partial observability); the residual will park on the latter indefinitely. Δr_k^{hist} is the signed change in this chunk’s disagreement across rescoring cycles. In from day one, not as an ablation.

Why score differences beat point-prediction variance. Contact dynamics are multimodal: slip versus stick, tip left versus tip right. If the true conditional is bimodal and *all* members correctly predict bimodal, variance over point predictions reads as high disagreement — a false positive, systematically largest exactly at contact. Score differences at matched noise level compare whole distributions, so agreement on a bimodal outcome correctly reads as agreement.

What this measures, and its blind spot. Note the absence of ground truth in (11): it is model-versus-model. It measures *what the data has not determined*, not how wrong the model is. Consequently, if all members are wrong *in the same way* — shared architecture bias, capacity limit, partial observability — they agree, and $r \rightarrow 0$ while error stays high:

$$r \rightarrow 0 \implies \text{more data will not help} \not\Rightarrow \text{the model is correct.} \quad (12)$$

This is a feature for exploration and a trap for evaluation.

Relationship to Distribution Matching Distillation. We borrow DMD’s *structure* — one KL expressed as a squared difference of two velocity/score fields — but not its *pairing*:

	Compares	Measures
DMD	s_{real} vs. s_{fake}	generator’s distribution vs. the data’s (model error)
Ours	v_{ϕ_i} vs. $\bar{v}$ (ensemble)	members vs. each other (epistemic uncertainty)

The choice is deliberate. DMD’s real-versus-fake pairing requires ground truth, which places it in the realized-regret family and reinstates the aleatoric and misspecification terms of (5). Members-versus-mean requires no ground truth and can therefore be evaluated on *imagined* states — which is what makes §4.5 possible at all. What is borrowed directly: the diffusion-KL identity, the adaptive w_σ normalization, and the σ -range restriction ($T_{\min} = 0.02T$, $T_{\max} = 0.98T$).

DMD’s literal pairing remains available as a *second* reward channel measuring model error rather than uncertainty — cheap once the σ -loop exists — but it is not a substitute, and mixing the two into one scalar without deciding which is intended would reproduce exactly the confound of (5).

One approximation we own. The mixture score is not the mean of the member scores: $\nabla \log \frac{1}{N} \sum_i p_i \neq \frac{1}{N} \sum_i \nabla \log p_i$, since the true mixture score is density-weighted. Lacking densities we use $\bar{v} = \frac{1}{N} \sum_i v_{\phi_i}$, which is exact to first order when members are close — as they are, being trained on the same data. We state the approximation rather than bury it.

4.3 The residual policy

Flow-matching policies have no tractable density, so we condition on the *sampled* base action and optimize only the residual:

$$\pi_\rho(a \mid s, a_b) = \mathcal{N}(a_b + \mu_\rho(s, a_b), \sigma_\rho^2 I), \quad \pi_{\text{ref}} = \mathcal{N}(a_b, \sigma_0^2 I), \quad (13)$$

$$\text{KL} = \frac{\|\mu_\rho\|^2}{2\sigma_0^2} + \frac{d}{2} \left(\frac{\sigma_\rho^2}{\sigma_0^2} - 1 - \log \frac{\sigma_\rho^2}{\sigma_0^2} \right). \quad (14)$$

Closed form; the KL ball is literally a ball around the base action. σ_0 is *measured* from the base policy’s own action variance across seeds, not tuned. The base policy’s stochasticity is not penalized, which is correct: it is part of the behavior we want to stay near.

Chunk-level actions. One Δ_k per chunk, held with a raised-cosine profile:

$$\Delta_k \sim \mathcal{N}(\mu_\rho(s_k, a_b, \text{phase}_k), \sigma_\rho^2), \quad a_t = \text{clip}(a_b(t) + w(t) \Delta_k, -1, 1). \quad (15)$$

Three reasons: (i) i.i.d. per-step noise averages out over the object’s response time and perturbs nothing; (ii) the reward is per-chunk, so action and reward share granularity; (iii) the RL horizon becomes 7 rather than 200. We expect (iii) to be the difference between the policy training and not.

4.4 Learning the residual in closed form

For a Gaussian residual, $\nabla_\mu \log \pi_\rho = (\Delta - \mu)/\sigma_\rho^2$, so the policy-gradient update is an advantage-weighted average of sampled perturbations,

$$\mu_\rho(s_k) \leftarrow \mu_\rho(s_k) + \eta A_k \frac{\Delta_k - \mu_\rho(s_k)}{\sigma_\rho^2}. \quad (16)$$

Given this structure, PPO is unnecessary. The KL-constrained improvement problem $\max_\pi \mathbb{E}_\pi[A] - \frac{1}{\beta} \text{KL}(\pi \parallel \pi_b)$ has the closed-form solution $\pi^*(a \mid s) \propto \pi_b(a \mid s) e^{\beta A(s, a)}$, fit by weighted regression:

$$\mu_\rho \leftarrow \arg \min_\rho \sum_j \text{softmax}_j(\beta A_j) \|\Delta_j - \mu_\rho(s_j)\|^2. \quad (17)$$

No clipping, no GAE, no importance ratio, and the trust region is *exact* rather than the crude surrogate that PPO clipping provides. We start here and escalate to PPO only if it plateaus.

4.5 The loop

Algorithm 1 Designed Interaction

```

1:  $\phi \leftarrow \text{TRAINWM}(\mathcal{D}_{\text{demo}})$ ;  $\{\phi_i\} \leftarrow N=5$  members, independent seeds;  $\pi_b \leftarrow \text{BC}(\mathcal{D}_{\text{demo}})$ 
2:  $\rho \leftarrow 0$ ;  $\mathcal{B} \leftarrow \mathcal{D}_{\text{demo}}$ 
3: for  $t = 1, \dots, T$  rounds do
4:   (1) Residual — imagination only, zero environment interaction
5:      $\triangleright$  diffusion WM: needs a distilled rollout engine, §4.9; otherwise use myopic greedy
6:     freeze  $\phi, \{\phi_i\}$ 
7:     for  $n_\rho$  iterations do
8:        $z_0 \sim \mathcal{B}$   $\triangleright$  cached latent
9:       for  $k = 0, \dots, 6$  do
10:         $a_b \leftarrow \pi_b.\text{SAMPLE}(s_k)$ ;  $\Delta_k \sim \mathcal{N}(\mu_\rho(s_k, a_b, \text{phase}_k), \sigma_\rho^2)$ 
11:         $r_k \leftarrow \text{Eq. (11)}$ ;  $z_{k+1} \leftarrow \text{TRUNK}(z_k, a_b + \Delta_k)$   $\triangleright$  1 fwd, no decode
12:      end for
13:       $\tilde{r} \leftarrow (r - \bar{r}_{\mathcal{B}}) / \text{std}_{\mathcal{B}}(r)$   $\triangleright$  renormalize per round
14:       $\rho \leftarrow \text{Eq. (17)}$  on  $A_k$  from  $\tilde{r}$ ,  $\gamma=0.9$ ;  $V_\rho$  reinitialized each round
15:    end for
16:    (2) Act — the only expensive step
17:    for  $B_{\text{ep}}$  episodes do
18:       $\tau, \text{states} \leftarrow \text{ROLLOUT}(\text{env}, \pi_b + \Delta_\rho)$ ;  $\mathcal{B} \leftarrow \mathcal{B} \cup \tau$ 
19:       $k^* \leftarrow \arg \max_k V_\rho(s_k)$ 
20:      for  $M$  branches do  $\triangleright$  simulator only
21:         $\text{SETSTATE}(\text{states}[k^*])$ ;  $\mathcal{B} \leftarrow \mathcal{B} \cup \text{ROLLOUT}(\text{env}, \pi_b + \Delta_\rho, 2 \text{ chunks})$ 
22:      end for
23:    end for
24:    (3) Update world model — weight, never select
25:     $\text{RESCOREBUFFER}(\mathcal{B}, \phi, \{\phi_i\})$   $\triangleright r_k$  is stale after every  $\phi$  update
26:     $w_k \leftarrow (r_k + \epsilon)^\alpha$ 
27:    for  $n_{\text{wm}}$  iterations do  $\triangleright n_{\text{wm}} : n_\rho \approx 5 : 1$ 
28:      batch  $\leftarrow (1 - r_{\text{mix}}) \text{SAMPLE}(\mathcal{D}_{\text{demo}}) + r_{\text{mix}} \text{SAMPLE}(\mathcal{B}, p \propto w_k)$ 
29:       $\phi \leftarrow \phi - \nabla_\phi \sum \ell(\phi; \text{batch})$ 
30:    end for
31:     $\{\phi_i\} \leftarrow$  refit on bootstrap resamples of  $\mathcal{B} \cup \mathcal{D}_{\text{demo}}$ 
32:    (4) Diagnostics
33:     $G_t \leftarrow \max_k r_k$  over imagined rollouts; phase-mix $t$ ; cf-err $t$ 
34:    if  $G_t < \tau$  then break
35:  end for

```

4.6 Data: collect $\neq$ store $\neq$ train on

Budget is spent only on *collection*; retained data is free forever.

	costs budget?	reaching, after solved	mechanism
collect new	yes	stop	$r_k \rightarrow 0 \Rightarrow \mu_\rho \rightarrow 0, V \rightarrow 0$
keep in buffer	no	keep	r_{mix} passive mixture
train on	compute only	keep, downweighted	$w_k \propto r_k^\alpha$

Dropping solved data causes forgetting; the target is *no new* reaching data, not *no* reaching data. **No phase detector is written anywhere:** uncertainty is the detector, and it retargets itself as the model improves.

Weight, do not select. Soft sampling $p \propto w_k$, never top- K . Top- K over a polytope collapses to near-duplicates — the failure of §2.5. Hard selection earns its place only when skipping render/encode saves real wall-clock (Stage 2).

Branching. To reach a contact state one must traverse the approach; branching pays that traversal once and amortizes it. With 7 chunks/episode and ≈ 2 contact chunks: $M=8$ branches at $\arg \max_k V_\rho$ gives ≈ 21 chunks with 16 contact ($3\times$ cost, $8\times$ contact data; contact share $29\% \rightarrow 76\%$). This is the operational answer to “we do not need more reaching data.”

4.7 Convergence

Three separate questions.

Inner loop (fixed ϕ). Converges: (17) is monotone improvement on a fixed reward under a KL constraint, with bounded Δ .

Outer loop. No saddle-point guarantee (nonconvex–nonconcave), but there is a structural reason it terminates that does not apply to GANs:

The buffer is monotone. Data is never deleted, so once the data pins down the dynamics at (z, a) , no future action un-pins it: $\text{Var}_\phi(z, a)$ is non-increasing on the visited set.

The residual’s high-reward territory therefore shrinks monotonically and never regenerates. This is an exhaustion process, not an arms race. The right theoretical frame is greedy maximization of a submodular coverage objective — mutual information between observations and parameters is submodular under conditional independence given ϕ , which underwrites near-optimal Bayesian experimental design — though the adaptive, policy-parameterized version is not literally covered by those results, and we cite it as intuition rather than as a theorem.

Termination. $r \rightarrow 0$ over the reachable set $\Rightarrow \mu_\rho \rightarrow 0$ (the KL anchor pulls it home) $\Rightarrow$ collection reverts to π_b . The system turns itself off; no stopping criterion is needed. $G_t \rightarrow 0$ certifies saturation over $\mathcal{D}_\epsilon$ — not global optimality.

Stabilizers. Two-timescale ratio $n_{\text{wm}} : n_\rho \approx 5:1$ (the model must be the fast player, or the residual locks onto regions the model never fixes and G oscillates); per-round renormalization of r (the reward shrinks *globally*, so advantages vanish before the relative signal does); critic reinitialization (it was fit to the old r); bounded Δ and the KL anchor; the learning-progress term.

4.8 Three levels of exploration policy

Level	Mechanism	Justified when
0	demonstrations only	—
1	designed distribution over open-loop parameters	primitives reach informative designs
2	learned closed-loop residual Δ_ρ	informativeness is state-dependent

The base policy already solves reachability. We test Level 1 before Level 2, and the decisive measurement is whether $\|\mu_\rho(s)\|$ varies systematically with episode phase. If it is roughly constant, a bandit over open-loop parameters suffices and we should report that.

4.9 Multi-step diffusion world models

Our world model is a multi-step denoising diffusion (flow-matching) model, not a single-step autoregressive latent model. This affects the two halves of the method in opposite directions.

The score is unaffected — indeed it is native. Equation (11) evaluates velocity fields *at* noise levels; it never samples from the model, so no denoising chain is run. One forward pass per $(\sigma, \varepsilon, \text{member})$ triple. Our existing scorer already executes precisely this loop (4 noise levels $\times$ 3 seeds per chunk); the change is to retain the velocity vector instead of reducing to a scalar loss. Moreover, the score-difference formulation *only exists* for diffusion models: a deterministic latent predictor has no score to difference, and would force us back to variance over point predictions with the multimodality failure of §4.2.

Imagination is what breaks. A single imagined transition costs a full denoising chain (≈ 30 steps, $\approx 4.75\text{s}/\text{chunk}$ in our implementation), so one 7-chunk rollout is $\approx 33\text{s}$ and the thousands of rollouts required per residual update are infeasible by three orders of magnitude. This is not a tuning problem.

Three resolutions.

1. **Myopic greedy (no rollout).** Scoring candidate Δ_k requires the velocity field at z_k , not at z_{k+1} , so one-step selection needs *zero* denoising. Sample ≈ 8 candidate perturbations per chunk, score each ($2\sigma \times 1 \text{ seed} \times 3 \text{ members} = 48 \text{ forwards} \approx 8\text{s}$), execute the argmax. Overhead $\approx 56\text{s}/\text{episode}$ on top of the existing pipeline; feasible immediately. Cost: no long-horizon information seeking and no amortized policy.
2. **Distill a rollout engine.** DMD’s purpose is exactly to convert a multi-step diffusion model into a one- or two-step generator. Use the full model as the object being improved *and* as the scorer, and a DMD-distilled few-step copy as the imagination engine, refreshed at LoRA granularity each round. This is why comparable systems adopt consistency-model backbones: real-time rollout is the enabling property.
3. **Small surrogate for imagination.** Cheapest to build, but the residual then optimizes against a different model’s uncertainty than the one being improved. Preferred only if (2) is out of reach.

Why this is a contribution rather than a workaround. PROWL also uses a diffusion video world model, and that is precisely why its adversary must roll out in the *environment*: realized regret against ground-truth rollouts, with PPO on environment samples. Multi-step denoising forecloses imagination, and imagination is our stated advantage. Therefore:

Distillation is not an engineering workaround. It is the mechanism that makes imagination-based adversarial curricula possible for diffusion world models, and it is what separates this method from environment-rollout adversaries.

The same DMD machinery thus does two jobs: score differences supply the reward, and distribution matching supplies the rollout engine.

Staging consequence. Stage 1 runs resolution (1), which requires no distillation and tests whether the signal separates arms at all. Only if it does is resolution (2) worth building; comparing (2) against (1) then measures directly whether informativeness is state-dependent (§4.8) — if greedy ties the amortized policy, we report that.

5 Experiments

5.1 Arms (fixed *collection* budget)

Arm	Residual	Purpose
demos only	$\Delta = 0$	floor
play / scripted-random	unconstrained scripted	prior-work protocol
random residual, matched $\mathbb{E}\ \Delta\$	$\mu_\rho \equiv 0$, same σ_ρ	decides the result
max-error curiosity	regret signal, top- K	our previous AL signal
reducible loss	$\ell - \ell_{\text{ref}}$	RHO-LOSS
top- K by r , no reweighting	—	ablation for §2.3(a)
unconstrained ($c_{kl}=0$)	—	$\approx$ Plan2Explore
PROWL-style regret	env rollouts per update	matched <i>env-step</i> budget
ours	Eq. (11), KL-anchored	—

The matched-magnitude arm is one flag ($\mu_\rho \equiv 0$) and isolates the *direction* of perturbation from its size; magnitude is matched by construction rather than by tuning. If ours ties it, the honest finding is that bounded random perturbation suffices — useful, cheaper to deploy, and still reportable. The PROWL comparison must be at matched *environment-step* budget, since that is the resource our method claims to save.

5.2 Metrics and evaluation protocol

- **Primary:** counterfactual *object-region* error under held-out interaction parameters (§5.2.1), reported as **mean / P75 / P90 / P99**, never mean alone.
- **Secondary:** task score of a control policy trained on world-model rollouts. This requires no pose readout at all and is the headline number.
- **Diagnostics:** G_t ; budget share by phase; $\|\mu_\rho\|$ versus phase; σ_ρ ; fraction of Δ at the clip boundary.
- **Not used:** PSNR / SSIM over full frames. These sit inside the misspecification floor and produced our uninformative null.

5.2.1 Measuring object error from a pixel-space model

The world model predicts pixels, not poses. It does not need a pose head: **the pose is present in the predicted pixels**, and the readout is an *evaluation instrument* applied to the model’s output — never part of the model, never used at inference. Ground truth is read directly from the simulator (`sim.data.get_body_xpos`), already exposed by our pipeline as world-space box corners ($n_{\text{obj}}, F, 8, 3$). No off-the-shelf pose estimator is required, and using one would stack its error on top of the model’s.

We use **corners rather than 6-DoF pose**: corners live in $\mathbb{R}^3$, so the error is a mean Euclidean distance in centimetres — one interpretable scalar — whereas 6-DoF forces an arbitrary weighting between translation error and a geodesic rotation error.

Two readouts, staged.

	Object-region error (Stage 1)	Frozen corner probe (Stage 2)
new code	none — already implemented	≈ 50 lines, trained once
mechanism	compare predicted vs. GT frames <i>inside</i> the GT object mask	latent $\rightarrow$ MLP $\rightarrow$ 8 corners, GT-supervised on real data
units	pixel / latent	centimetres
decoding	only if masking in pixel space	none — runs in latent space
saturates	yes (5 cm and 50 cm displacement look alike once the object leaves the mask)	no

Stage 1 uses the object-region error because it exists today, requires no training, and is sufficient to *rank* arms. The probe is added when physical units are needed — specifically for the tail percentiles, since saturation is exactly what makes masked pixel error stop discriminating at P90/P99, which is where the mechanism of §5.3 is supposed to appear.

Three requirements on the probe. Each is a way to silently invalidate the experiment.

1. **Train once, freeze, share across all arms.** Fitting a probe per arm measures probe capacity rather than world-model quality and will make every arm look alike.
2. **Report the probe’s own floor** — its error on *real* held-out latents — as a horizontal line on every plot. An arm near that floor is not distinguishable from any other.
3. **Handle distribution shift.** The probe is fit on encoded *real* latents but applied to *predicted* latents, which drift off-manifold with horizon. Mitigate by including predictions from a fixed reference model in the probe’s training mix — never predictions from the arms themselves, which would reintroduce (1).

Sampling: the model outputs a draw, not a point prediction. A diffusion world model returns a *sample* from $p(\text{video} \mid \text{context}, \text{actions})$, so comparing one rollout against one ground-truth trajectory conflates wrong dynamics with a different mode. We draw $K=5$ rollouts per evaluation episode and report both

$$\min_k \|\hat{c}_k - c^{\text{GT}}\| \quad (\text{best-of-}K: \text{ mode coverage}), \quad \frac{1}{K} \sum_k \|\hat{c}_k - c^{\text{GT}}\| \quad (\text{mean: calibration}). \quad (18)$$

They fail in different ways and both are needed.

Protocol.

1. Hold out a slice of design space (e.g. push azimuth in $[\pi, 3\pi/2]$) from *all* arms’ training data.
2. Reset to a fixed seeded state; execute the held-out action sequence.
3. Ground truth: corners per frame from the simulator.

4. Model: condition on the first k context frames, then roll out **free-running** on the same action sequence. *Not teacher-forced* — teacher forcing feeds ground-truth latents back in and tests denoising rather than dynamics. (Our training-time scorer is teacher-forced, which is correct there and wrong here.)
5. Report error versus horizon h , and at the final h report mean / P75 / P90 / P99.

Real-robot caveat. None of this is free outside simulation: ground truth would require a pose estimator, markers, or motion capture, and would itself carry error. We therefore keep the primary quantitative claim in simulation and rely on the downstream policy task score — which needs no pose readout — for any real-robot result.

5.3 The signature that confirms the mechanism

Because the target (3) is a $\sup_u$ and not an $\mathbb{E}_u$:

Gains must be larger in the tail than in the mean.

Random data reduces the mean. PROWL’s held-out table shows exactly this fingerprint (−3.5% mean, −8.0% hardest quartile, −8.7% hardest decile). **If our gain is flat across percentiles, the mechanism is not why we won**, and that must be diagnosed before writing up.

5.4 Pre-registered predictions

1. On the old pool (no contact-exciting primitive), $G \approx 0$ and all arms tie — retrodicting §2.5.
2. Admitting a lateral-push primitive raises G sharply *with no new data*.
3. Ours > matched-random > curiosity on counterfactual object-region error.
4. Sweeping ϵ yields an **interior** optimum.
5. The no-reweighting ablation collapses to near-duplicate picks.
6. Reaching’s budget share decays monotonically across rounds.
7. $\|\mu_\rho(s)\|$ is larger near contact than during transport (justifies Level 2).

Predicting the ties is part of the claim; a framework that only predicts wins is not falsifiable.

5.5 Configuration

chunk horizon	7	γ	0.9
ensemble N	5, independent seeds	encoder	frozen
σ_0	measured from π_b	c_{kl}	sweep {0.5, 1.0, 1.5}
r_{mix} (passive)	0.5	α (weights)	1.0
branching M	8	$n_{\text{wm}} : n_\rho$	5:1
σ -grid	4 levels, sweep σ_{max}	noise draws	2 per level
episodes/round	20	rounds	≈ 6

Randomize initial object placement every episode. Without this the scene saturates (the ≈ 30 episode identities of §2.5) independent of everything above.

5.6 Failure signatures

Symptom	Diagnosis
G flat across rounds	correlated members, or chasing something unlearnable
G oscillates	residual outpacing model; raise n_{wm}
G plateaus above 0	parked on the unlearnable; strengthen progress term
$G \rightarrow 0$ but held-out error flat	ensemble collapsed — fake saturation
Δ pinned at clip bound	c_{kl} too low
$\sigma_\rho \rightarrow 0$	entropy lost to KL; residual went deterministic
reach share not decaying	reaching not actually solved, or members cannot tell

The fourth is the dangerous one: every other failure looks like failure, that one looks like success. Track G_t and counterfactual error on the same axes every round — they must fall together.

6 Limitations

1. **Necessity versus efficiency.** Infinite design gain requires a genuine range deficiency, testable only where ϕ has a low-dimensional structured component. Over projected network parameters we are entitled to a *conditioning* claim, not an identifiability claim.
2. **Locality.** r , V , and the reachable set all move as ϕ trains. Any convexity holds per round; no global optimality is claimed.
3. **The certificate is about data, never about quality.** $r \rightarrow 0$ means more data of this kind will not help; it does not mean the model is correct.
4. **Appearance confound.** If the leading disagreement directions load on background and lighting rather than object motion, the method chases visual novelty exactly as curiosity does. The object-token mask is a mitigation, not a proof; it must be verified directly.
5. **Branching is simulator-only** and breaks the occupancy interpretation — the training distribution is then a free design measure rather than one induced by a policy from ν_0 , which is mathematically cleaner but introduces a train/deploy shift that must be measured. If the pitch is real-robot data efficiency, branching is a sim-only ablation.
6. **We do not claim to teach “physics” specifically.** The algorithm targets epistemic uncertainty, which coincides with physics only insofar as physics is what the model is uncertain about. We will write *action-response fidelity* or *contact dynamics* in results and reserve “physics” for motivation.

7 Build Plan

Stage	What	Gate	Cost
0a	members on existing demos; per-token disagreement heatmap	lights up on object+gripper	1 day, no training
0b	re-score the existing null arms with a downstream metric	does the null survive?	1 day, no training
1	chunk-level residual + AWR in imagination; all arms; small pixel WM	ours > matched-random at P90	$\approx$ 2 weeks
2	frozen DINOv2 encoder: does the ranking survive an encoder?	either answer publishable	$\approx$ 1 week
3	fine-tune the large video model per arm; policy task score	ranking transfers	$\approx$ 2 weeks

Stages 0a and 0b run entirely against artifacts already on disk and must precede any new compute. 0a kills the appearance confound before it can invalidate downstream numbers; 0b determines whether the prior null was a metric artifact.

Positioning statement. *Adversarial world-model curricula require environment rollouts for every policy update, which is prohibitive for physical robots. We show the adversary can be optimized entirely in imagination using epistemic score disagreement, achieving comparable model improvement at a fraction of the interaction budget — and that the resulting model trains better control policies.* PROWL is the baseline, not the competitor.